

Renke (Richard) Han | PhD

Personal Webpage – <https://renke-richard-han.github.io>

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Profile

Profound knowledge about global xEV market trends (Powertrain, Zonal, EEA) at different regions. 11 years cutting-edge experiences in power electronics design with both software and hardware. Driving crossing function collaboration with multi-culture background. Strong business oriented mindset with strong system understanding:

- Working and collaborating with multi culture teams (Asian, European, African, US) for 11 years
- As OBC/DCDC/X-in-1 application owner reporting directly to head of product line with above 1B revenue
- Clear view about OEM/Tier 1s development trends and requirements for xEV, and being strongly involved in customer RFIs for powertrain platform, E/E architecture
- Profound understanding for microcontroller, power device, and system architecture evaluation paths
- Embedded firmware development: Digital controller design/implementation/validation with AURIX, STM32@STMicroelectronics, XMC@Infineon, C8051F352@Silicon lab
- CAD and simulation tools: Altium, LTspice, PLECS, SIMULINK/Matlab

Professional Experiences

Staff -> Senior Staff -> Principal Engineer - AURIX System Architect <i>Automotive(ATV) · Infineon Technologies AG · Germany</i>	2023.06 – present
Senior -> Staff System Engineer <i>Green Industrial Power (GIP) · Infineon Technologies AG · Germany</i>	2021.08 – 2023.05
Postdoctoral Research Associate <i>Department of Engineering Science · University of Oxford · United Kingdom</i>	2018.11 – 2021.07
Departmental Tutorship for Semiconductor Devices <i>Department of Engineering Science · University of Oxford · United Kingdom</i>	2020.10 – 2021.07
Visiting Scholar <i>Automatic Control Laboratory · École Polytechnique Fédérale de Lausanne · Switzerland</i>	2017.02 – 2017.08

Education

Ph.D. in Power Electronics Systems <i>Department of Energy Technology · Aalborg University · Denmark</i>	2015.11 – 2018.10
MA.Sc in Control Theory & Control Engineering <i>GPA: 90/100, Ranking: 1/134</i> <i>School of Information Science & Engineering · Northeastern University · China</i>	2013.09 – 2015.07
B.Eng in Automation <i>GPA: 88.8/100, Ranking: 14/270</i> <i>School of Information Science & Engineering · Northeastern University · China</i>	2009.09 – 2013.07

Experiences

System architect reporting directly to Head of AURIX Product Line above 1B revenue <i>xEV application: On-board charger + HV/LV DCDC, traction inverter</i>	2023.06 – present
◦ Next generation automotive MCU definition - Peripheral KPI definition including PWM resolution and special driving signal (eg: synchronous rectifier, single stage OBC driving signal for bidirectional switches), ADC sampling rate and accuracy, fast comparator and DAC (peak current mode control), bus architecture, core performance for real-time application - Competitor products features comparison (TI, Renesas, NXP) and fighting guides generation ◦ Strongly customer engagement in EMEA, JP and GC (on-site technical promotion, tech-day, RFI response) - Success G2M strategies (theoretical/simulation proof) for AURIX as solid enough product with system benefit - System level BOM cost optimization for customers (different scenarios comparison) ◦ As a technical stakeholder and project manager leading innovation projects (20 people projects) - Real time junction temperature estimation for traction inverter using AURIX TC4xx parallel process unit - New OBC + DCDC architecture by using one MCU, validation through hardware-in-loop setup - Firmware architect for 3.7kW single stage OBC - White paper generation (Click here), Infineon webinar (Click here)	
Industrial Reference Design and Digital Service <i>Industrial application</i>	2021.08 – 2023.05

- 22kW General Purpose Motor Drive (Click here)
- IGBT power module real-time status monitor
 - Thermal impedance fitting, digitization, uC implementation on XMC and associated tool suite development on Matlab
 - Firmware development for power loss calculation with Junction temperature calculation embedded with motor control
- Flyback auxiliary power supply design (100 W, input DC link voltage 650V, output voltage: 15V and 24V)
- Signal chain design print circuit board design for several applications: T-sense module, Vce saturation measurement

Project: Robust Extra Low Cost Nano-Grids for developing world

Funded by UK Engineering and Physical Sciences Research Council (EPSRC)

2018.08 – 2021.07

- Develop scalable 12-200V, 300W multi-port isolated DC-DC power converters for PV-battery system
 - Topology and passive components selection, high frequency magnetic components design;
 - Embedded coding including communication(UART/I2C/SPI), measurement(ADC, Op-amp), digital control, Li-ion battery charge/discharge, cells balancing and protection strategies on STM32;
 - Construct Nanogrid prototype demonstration with converters, ten 240Wh li-ion batteries, 7.2kWh lead-acid batteries;
 - Design qualification control tests including measurement accuracy, safety, reliability;
- Coordinate and communicate with industrial (Tropical Power Cooperation) and academic (Cardiff University) partners
- On-the-ground experience deploying 10 designed converters to one village in Africa providing normal life electricity

Project: Pure software solution for power line communication (UK patent)

Funded by University of Oxford and Oxford Innovation Institute

2020.06 – 2021.03

- Amplitude modulation achieving 2 kbps communication with only 8 kHz sampling frequency using the frequency band above control bandwidth but below sampling frequency
- Pure software modulation and demodulation with Manchester Encoding

Consultant Project: 5kW inductive Charger for Electrical Vehicle

Funded by University of Oxford

2020.09 – 2020.11

- Analysis and simulation for weak coupling transformer and its equivalent circuit
- Characteristics analysis about tuned LCL circuit including LCL load resonant converter and secondary pickup circuit
- Calculation about max current through transformer winding and max power transfer capacity providing guideline for practical design

Awards and Prizes

Representative (1/7) of University of Oxford , Global Young Scientist Summit ·Singapore	2021.01
IEEE ECCE Travel Grants ·Cincinnati ·US	2017.10
Best Oral Presentation at IEEE 42nd IECON ·Florence ·Italy	2016.11
Outstanding Master Thesis at Provincial-level ·China	2015.06
Outstanding National Scholarship ·China	2014.06

Other Experiences

Member of the IEEE Power Electron., and Ind. Electron. Societies	2015.11–
President of Chinese Students and Scholars Association, Denmark	2016.02 - 2017.08

Skills

Proficient: Altium Designer, PLECS, C language, AURIX, TI C2000, STM32F103, LTspice, Eagle
 Language: Chinese (Native), English (Business Level), German (B1)

Selected Top Publications

1 UK patent, 15 Journal Papers (5 as first author in IEEE Transactions on Power Electronics, Power Systems, Industry Application), 10 Conference Papers (oral presentations)

Google Scholar Citations: 2128, H-index: 16 i10-index: 20

- 1 **R. Han** and D. J. Rogers, "Zero-additional-hardware power line communication for dc-dc converters," *IEEE Transactions on Power Electronics*, vol. 37, no. 11, pp. 13 107–13 118, 2022.
- 2 **R. Han**, L. Meng, J. M. Guerrero, and etc, "Distributed nonlinear control with event-triggered communication to achieve current-sharing and voltage regulation in dc microgrids," *IEEE Transactions on Power Electronics*, vol. 33, no. 7, pp. 6416–6433, 2018.
- 3 E. Shelton, **R. Han**, D. Rogers, J. Carter, L. Louco, M. Beadman, and P. Palmer, "Comparison of fast switching high current power devices," in *PCIM Europe digital days 2021; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, 2021, pp. 1–8.